

Detection and severity analysis of tea leaf blight based on deep learning

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ARTICLE INFO

Keywords:

Tea leaf blight
Disease detection
Severity analysis
Deep learning

ABSTRACT

At present, the detection and severity estimation of tea diseases mainly rely on manual methods, which are time consuming and laborious. Existing machine learning and image processing methods used in disease detection and severity analysis of tea leaf blight (TLB) images captured in natural scenes have low accuracy because of the influence of light variation, shadow, varying shapes, and mutual occlusion of leaves. The current study proposes a deep learning method to improve the performance of detection and severity analysis of TLB. A Retinex algorithm is utilized to enhance the original images and reduce the influence of light variation and shadow. The TLB leaves are detected using a deep learning framework called Faster Region-based Convolutional Neural Networks, to improve the detection performance of blurred, occluded, and small pieces of diseased leaves. The detected TLB leaves are inputted into the trained VGG16 networks to achieve severity grading and facilitate disease severity analysis. Experimental results show that the detection average precision and the severity grading accuracy of the proposed method are improved by more than 6% and 9%, respectively, compared with the classical machine learning methods.

1. Introduction

Tea is an important economic crop and is welcomed by many people because of its health care function. Therefore, the scale of tea cultivation has increased rapidly in many countries [1]. In 2018, the tea plantations in the major tea-producing provinces of China covered an area of 2.93 million hectares, with an output of 2.616 million tons of tea, and the total output value of tea exceeded 200 billion yuan. Disease infection is an important factor that affects tea yield and quality. Tea leaf blight (TLB) is a common tea disease in China, which is distributed in all tea-producing areas in the country. The diseased leaves frequently drop prematurely, and the new shoots become withered when a tea tree is infected, thereby resulting in the weakness of tree vigor. The tea garden presents a brownish color, and young tea trees can die when the disease is serious, thereby causing immense losses to tea farmers [2]. The accurate detection of TLB and analysis of disease severity are important to effectively prevent and control TLB for ensuring tea quality and increasing the income of tea farmers.

At present, the detection and severity analysis of TLB mainly rely on manual methods, which require plant protection experts to go to the field to identify the disease and determine its severity. Many tea gardens are located in rugged mountain areas, and manual field

This paper is for regular issues of CAEE. Reviews processed and approved for publication by the co-Editor-in-Chief Huimin Lu

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<https://doi.org/10.1016/j.compeleceng.2021.107023>

Received 24 January 2020; Received in revised form 14 January 2021; Accepted 30 January 2021

Available online 11 February 2021

0045-7906/© 2021 Published by Elsevier Ltd.

investigation is time consuming, laborious, and expensive. Thus, with the development of computer technology, machine learning and image processing methods that do not require manual intervention have been widely used in the detection and identification of plant diseases [3–7]. The classical machine learning methods of plant disease identification and detection require manual feature extraction of diseased leaves. The accuracy of manual features has a substantial influence on the identification and detection performance of plant diseased leaves because their color, texture, and other features are similar [8].

Prevention and control of TLB disease are urgently needed. This study aims to use deep learning methods to solve the multi-task processing problems of accurate detection of TLB and analysis of disease severity in natural scenes. The results can help tea farmers to choose and use pesticides reasonably, realize the effective prevention and control of TLB disease, and reduce the pollution of tea and soils simultaneously.

The main contributions of this study are listed as follows:

- 1) A deep learning-based method for detecting TLB and analyzing the severity of TLB is proposed.
- 2) An improved Faster Region-based Convolutional Neural Networks (Faster R-CNN) model is used to detect TLB, which can improve the effect of multiscale detection.
- 3) A VGG16 deep network is used to accurately classify the severity of TLB.
- 4) The Retinex algorithm is used to enhance images and reduce the influence of light variation and shadow.
- 5) This method provides a basis for the automatic prevention and control of TLB and is conducive to the rational use of pesticides.

The rest of this paper is organized as follows. Section 2 provides an overview of related works. Section 3 describes the data acquisition and preprocessing. Section 4 gives a detailed description of the model, design, and goals for TLB detection and severity analysis. Section 5 provides experimental results and analysis. Section 6 presents the discussions. Finally, Section 7 concludes and recommends future work.

2. Related works

Convolutional neural network (CNN) models are commonly used particularly in medical image analysis [9–13] due to the success behind their formulations [14] with appropriate parameters [15]. Although, there are challenging issues about their implementations, alternatives solutions have been proposed [16]. These models were inspired by biological and natural visual cognition mechanisms. They have advanced semantic expression ability and strong feature robustness. Table 1 shows some applications of CNN in plant disease identification. Compared with classical machine learning methods, deep learning methods have higher accuracy in identifying plant diseases and stronger transfer learning ability. However, the latter generally require considerable images to train models, and the collection of images has certain illumination requirements [24].

CNN models, such as VGGNet, ResNet, AlexNet, GoogLeNet, Faster R-CNN, and SSD, have also been applied in plant disease detection and severity estimation. Mohanty et al. analyzed the performance of AlexNet and GoogLeNet for image-based plant disease detection [25]. Ozguven et al. used a Faster R-CNN model to estimate the severity of diseases in sugar beet leaves [26]. Wang et al. compared the performance of a series of deep CNN to estimate the severity of the apple black rot [27]. Liang et al. proposed a plant disease severity estimation network PD²SENet, which achieves excellent comprehensive performances [28]. Few studies use the deep learning model for tea disease detection and severity analysis. The performance of detection and severity estimation directly using the above deep learning methods is not good given that the tea disease images taken in natural scenes are affected by light variation, shadow, and complex background.

The current study proposes an image-based method to detect and analyze TLB using an improved Faster R-CNN model. In this method, a Retinex algorithm is used to enhance the images and reduce the influence of light variation and shadow on the detection performance. An improved Faster R-CNN model is used to detect TLB, and a VGG16-based disease severity grading model is utilized to grade the detected diseased leaves and analyze the overall disease severity of TLB. The proposed method is suitable for TLB images captured in natural scenes and overcomes the influence of complex background environment, light variation, leaf occlusion, and inconsistent scale on the detection and disease analysis of TLB.

Table 1
Some applications of CNN in plant disease identification.

Plant disease	Model	Image acquisition	Reference
Tea diseases	VGG16	Captured in tea gardens	Hu et al. [8]
Rice diseases	Deep CNN	Captured in rice experimental field	Lu et al. [17]
Wheat disease	ResNet	Captured in two pilot sites	Picon et al. [18]
Diseases of 25 plants	AlexNet et al.	PlantVillage	Ferentinos [19]
Diseases of 5 plants	Deep CNN	Downloaded from the Internet	Sladojevic et al. [20]
Tea diseases	LeafNet	Captured in natural environments	Chen et al. [21]
Tea diseases	Concatenated CNN	Collected from plantations	Krisnandi et al. [22]
Tea diseases	Improved deep CNN	Captured in tea gardens	Hu et al. [23]

3. Image acquisition and preprocessing

The TLB images used in the current study were captured in Tianjingshan National Forest Park and the Tea Garden of Miaozhu Village. The former is located in the south of Hefei, capital of Anhui Province, China (31°14'N, 117°36'E, 40 m a.s.l.). The latter is located in Chizhou City, Anhui Province, China (30°03'N, 117°09'E, 96 m a.s.l.). The image acquisition device was a Canon EOS 80D handheld camera. All images were captured in the field, and a total of 398 TLB images were captured. Fig. 1 shows the image acquisition device and one captured TLB image. Diseased leaves in TLB images and their severity were annotated by several agriculturalists from the National Engineering Research Center for Agro-Ecological Big Data Analysis & Application using the image annotation tool LabelImage.

In the diseased leaf detection experiment, the TLB images are randomly divided into a training and a test set according to a ratio of 4:1. The training set contains 318 TLB images, whereas the test set contains 80 TLB images, as shown in Table 2.

In the disease severity analysis experiment, the data set is the diseased leaf images cut out from the images in the training set of the diseased leaf detection experiment. Table 3 shows the number of diseased leaf images used in the disease severity analysis experiment. In Table 3, the number of images of the mildly diseased leaf in the training set is augmented to 700 through symmetrical transformation and rotation to keep the training set balance. Fig. 2 shows a mildly and a severely diseased leaves.

Although CNN can deal with light variation, training a CNN model needs a lot of data. The number of images of the mildly diseased leaf in the training set is only 700 after augmentation. Therefore, the number of training samples is still insufficient. Considering that all images were captured in natural scenes, the current study uses the Retinex algorithm to enhance the images and reduce the influence of light variation and shadow [29].

Retinex algorithm assumes that the original image $S(x, y)$ is the product of the reflected image $R(x, y)$ and the illuminated image $L(x, y)$, and can be expressed as follows:

$$S(x, y) = R(x, y) \times L(x, y), \quad (1)$$

The reflected image $R(x, y)$ carries the image details. The reflected image $R(x, y)$ can be obtained by removing the illuminated image $L(x, y)$ from the original image $S(x, y)$:

$$\log[R(x, y)] = \log[S(x, y)] - \log[L(x, y)], \quad (2)$$

The illuminated image $L(x, y)$ can be approximated by the convolution of the original one $S(x, y)$ and a Gaussian kernel $G(x, y)$, so:

$$\log[R(x, y)] = \log[S(x, y)] - \log[(S(x, y) * G(x, y))]. \quad (3)$$

The pixel values of $R(x, y)$ are mapped to [0,255], and the enhanced image is obtained. Fig. 3 shows the original image, and the enhanced image obtained using the Retinex algorithm.

4. Model, design, and goals

4.1. Diseased leaves detection

Faster R-CNN was proposed on the basis of R-CNN and Fast R-CNN [30-32]. Faster R-CNN directly uses region proposal network (RPN) to generate a detection box instead of the traditional sliding-window method, which immensely improves the generating speed of the detection box. Faster R-CNN is used to detect TLB because of its excellent detection performance. However, the images used in the current study were captured in natural scenes. The diseased leaves in the images had different shapes and large angle variations, and the leaves covered one another. Therefore, the performance of the original Faster R-CNN in detecting TLB leaves is not good. The current study proposes an improved Faster R-CNN model to detect TLB and improve detection performance. Fig. 4 depicts the structure



(a) Image acquisition device (b) One captured TLB image

Fig. 1. Image acquisition device and one captured TLB image.

(a) Image acquisition device; (b) One captured TLB image.

Table 2
Number of TLB images used in the diseased leaf detection experiment.

Data set	Training set	Test set
Number of images	318	80

Table 3
Number of diseased leaf images used in the disease severity analysis experiment.

Data set		Training set (original)	Training set (augmented)	Test set
Number of images	Mild	270	700	100
	Severe	700	700	100

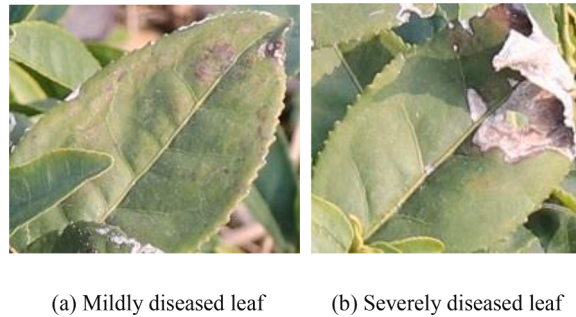


Fig. 2. A mildly and a severely diseased leaves.
(a) Mildly diseased leaf; (b) Severely diseased leaf.

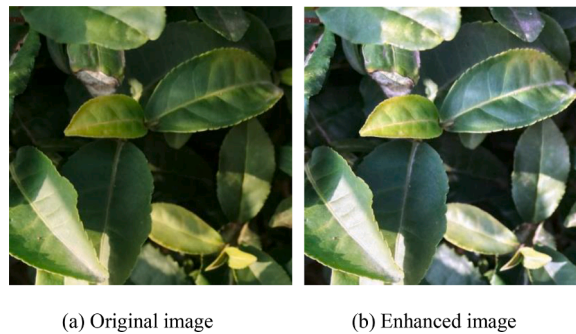


Fig. 3. An original image and the enhanced image obtained using the Retinex algorithm.
(a) Original image; (b) Enhanced image.

of the improved Faster R-CNN model, which mainly includes Feature Pyramid Networks (FPN) and pyramid Region Proposal Network (PRPN). Fig. 5 shows the structure represented by the green box in Fig. 4.

4.1.1. Feature pyramid network

The RPN and region of interest (RoI) of the original Faster R-CNN model are used on the last layer of the feature map output by the feature extraction network when used for object detection. Hence, the features of other layers are ignored, and some details are lost. However, these details can improve detection performance to a certain extent. FPN can be used for multiscale feature extraction and fusion [33]. Using FPN to extract and fuse multiscale detailed features of diseased leaves can improve the detection performance because of the complex background of TLB images captured in natural scenes and the curling and coloboma of diseased leaves. Fig. 6 shows the multiscale feature extraction and fusion by FPN. In Fig. 6, the blue boxes are the multiscale feature maps extracted by ResNet-101, and the brown ones are the fused feature maps (FFM).

The current study uses ResNet-101 as the backbone networks to build FPN [34]. Traditional CNNs or fully connected networks may have information losses during information transmission and may cause gradient disappearance or explosion problems. ResNet maintains the integrity of information by directly bypassing the input information to the output. The entire networks only need to learn the differences between the input and output, which will accelerate the training of the neural networks and improve model

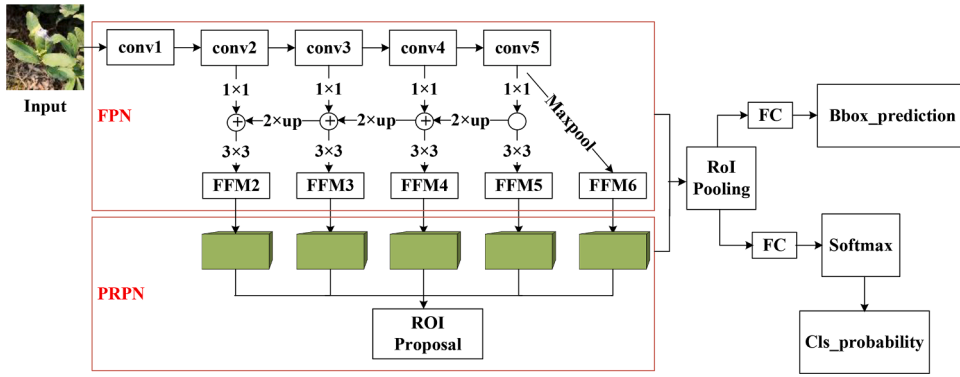


Fig. 4. Schematic of the improved Faster R-CNN model structure.

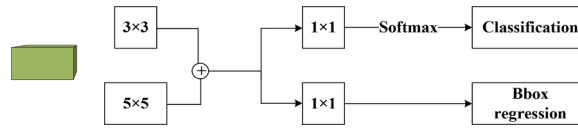


Fig. 5. Structure represented by the green box in Fig. 4.

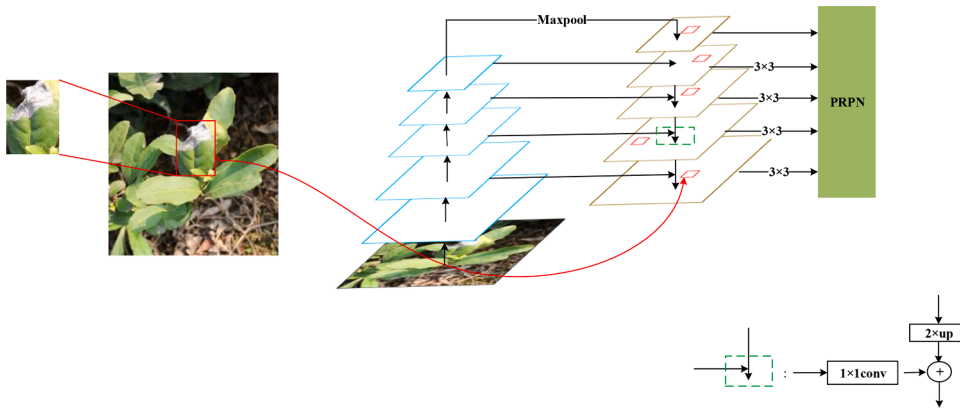


Fig. 6. Diagram of multiscale feature extraction and fusion by FPN.

performance. The last fully connected and pooled layers in the ResNet-101 networks are removed to facilitate the extraction of the features of diseased leaves.

4.1.2. Pyramid region proposal network

The PRPN inputs are the FFM generated by FPN, and the output is a rectangular box collection of object candidate regions. The original RPN directly extracts the candidate regions using a single 3×3 convolution to slide on the feature map output by FPN. Each pixel of the feature map is mapped back to the corresponding coordinate point of the original image through the 3×3 convolution. Three ratios and three scales are generated, and nine candidate regions (anchors) with different sizes are generated using this point as the center.

The scale of the TLB leaves immensely varies because of the distance and angle during shooting. Receptive fields with various sizes can be used to obtain remarkable candidate regions of diseased leaves with different scales. The current study proposes a PRPN structure. This structure uses two different sizes of sliding windows to convolve the feature map output by FPN to generate great candidate regions and improve the detection effect.

The anchor of the original RPN uses three basic scales of length–width ratio of 0.5, 1, and 2, which are set to the Visual Object Classes dataset [35]. The shapes of the diseased leaves of TLB images captured in natural scenes are diverse. Disease may occur on small tender or large mature leaves, and the shapes of diseased leaves will change with different disease severities, such as leaf curl and necrotic area drop. The three original basic ratios can no longer meet the requirements for accurate detection of diseased leaves. The current study used the K-means clustering algorithm to calculate the length–width ratio of diseased leaves of TLB images. Eight kinds of

length–width ratios (0.5, 0.68, 0.8, 1, 1.3, 1.44, 1.55, and 2) were obtained, which were consistent with those of tea leaves with different severities of disease and occlusion.

4.1.3. Detection procedures

The training data of TLB are inputted into the FPN to extract the features. Then, the FFM outputted by FPN are sent to the PRPN to generate candidate regions. The features of the candidate regions are processed into fixed-size feature vectors by the RoI pooling layer. Subsequently, such features are sent to the fully connected layer to output the specific classifications and their probabilities and bounding box predictions. During the testing process, the preprocessed test data of TLB are inputted into the trained Faster R-CNN, and the diseased leaf detection boxes are outputted.

The specific steps of the algorithm are described as follows:

Step 1. TLB images used in the diseased leaf detection experiment are enhanced and randomly divided into a training set and a test set with a ratio of 4:1;

Step 2. The images in the training set are inputted into the FPN. The multiscale features of the training data are extracted and fused by FPN to generate five-layer feature maps, namely, FFM2–FFM6 (Fig. 4);

Step 3. The FFM are inputted into the PRPN to generate the anchors using 3×3 and 5×5 convolutions;

Step 4. Softmax is used to classify the generated anchors to obtain the foreground and background;

Step 5. The offset of bounding box regression of anchors is calculated to determine the accurate position of the proposal boxes;

Step 6. The original feature maps and proposal boxes with different sizes obtained by the PRPN are inputted into the RoI pooling layer. The background and diseased leaves are classified using the fully connected and Softmax layers to obtain precise boxes;

Step 7. The images in the test set are inputted into the trained FPN, and the features are extracted and fused through Step 2;

Step 8. The FFM of the test data are inputted into the trained PRPN to position the diseased leaves in the test image, and the detection boxes are drawn through Steps 3–7.

Table 4 shows the pseudocode of the diseased leaves detection algorithm.

4.2. Disease severity analysis

An accurate analysis of the disease severity of TLB can prevent the spread of disease caused by insufficient drug use and environmental pollution caused by excessive drug use. Nevertheless, this analysis is conducive to rational drug use. Diseased tea leaves or the entire tea tree can be cut to avoid the spread of severe diseases. In the current study, a VGG16 deep networks is used to construct a disease severity analysis model, which is used to extract the features of TLB and classify the severity of the disease.

4.2.1. VGG16 deep networks

VGG16 is a deep CNN developed by the Computer Vision Group of Oxford University and Google Research Institute [36]. The CNN is realized by stacking convolution kernels with the same size (e.g., 3×3) and pooling layer (e.g., 1×1). VGG16 is shallower and requires less resources to train compared with ResNet. Fig. 7 shows the specific structure of VGG16 deep networks.

The VGG16 limits the size of input images to 224×224 . However, the images of TLB in the current study are presented as rectangles with inconsistent aspect ratios. The diseased leaf images are padded into a square and resized to 224×224 before they are inputted into the networks to avoid structure information loss of input data caused by resizing. Fig. 8 shows examples of directly resizing and padding images.

Table 4

The pseudocode of the diseased leaves detection algorithm.

Algorithm 1	Diseased leaves detection
Input: TLB images	
Output: Diseased leaves detection results	
1: TLB images are enhanced by Retinex	
2: FFM2-FFM6 ← images in the training set are input into the FPN	
3: Anchors ← PRPN extracts candidate regions on FFM2-FFM6 and calculates the offset of bounding box regression	
4: Diseased leaves with bounding boxes ← anchors and FFM2-FFM6 are input into RoI pooling layer, fully connected, and Softmax layers	
5: Trained detection model ← save weights	
6: Results ← TLB images in the test set are inputted into the trained model	

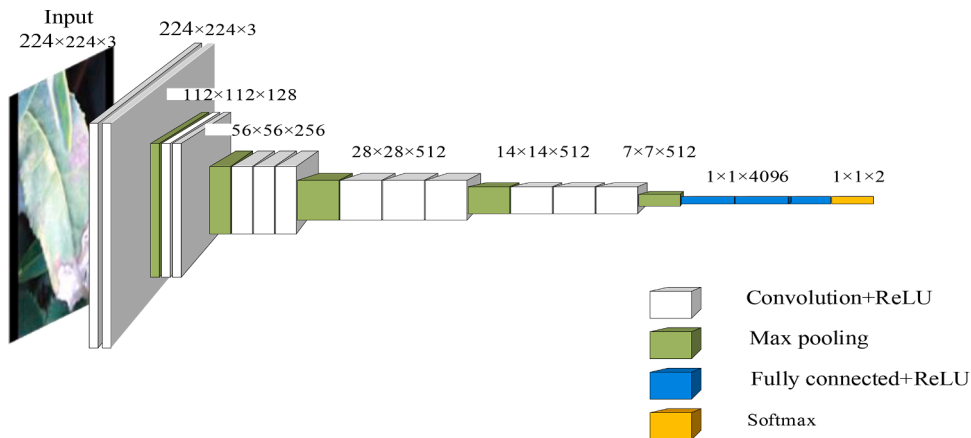


Fig. 7. Structure of VGG16 deep networks.

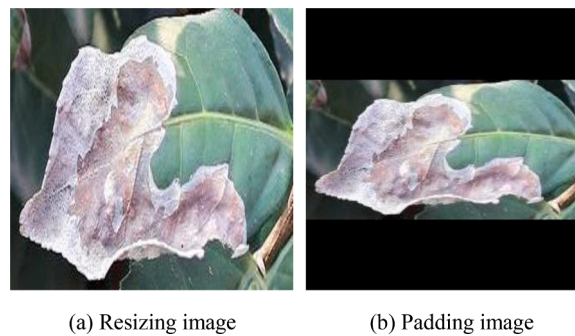


Fig. 8. Examples of resizing and padding images.

(a) Resizing image; (b) Padding image.

4.2.2. Disease severity analysis procedures

The specific steps of the disease severity analysis are described as follows:

- Step 1.** The diseased leaf images are obtained by cut out from the images in the training set of the diseased leaf detection experiment;
- Step 2.** The number of images of the mildly diseased leaf in the training set is augmented to 700 through symmetrical transformation and rotation;
- Step 3.** Diseased leaf images in the augmented training set are padded into a square and resized to 224×224 ;
- Step 4.** The padded and resized images are inputted into the VGG16 deep networks model, and the Stochastic Gradient Descent (SGD) algorithm is used to optimized the model;
- Step 5.** The detected diseased tea leaves using the detection algorithm are padded and inputted into the trained VGG16 deep networks model to classify the severity of the disease;

Table 5

The pseudocode of the disease severity analysis algorithm.

Algorithm 2	Disease severity analysis
Input: Diseased leaf images	
Output: Disease severity analysis results	
1: The diseased leaf images are obtained by cut out from the images in the training set of the diseased leaf detection experiment	
2: The number of images of the mildly diseased leaf in the training set is augmented	
3: The images in the augmented training set are padded and resized to 224×224	
4: Trained VGG16 model $\leftarrow$ images in the training set are inputted into the VGG16 and the SGD algorithm is used to optimized the model	
5: Classification results $\leftarrow$ detected diseased tea leaves using the detection algorithm are padded and inputted into the trained VGG16 model	
6: Severity analysis results $\leftarrow$ classification results are counted	

Step 6. The classification results are counted to achieve disease severity analysis.

Table 5 shows the pseudocode of the disease severity analysis algorithm.

4.3. Flowchart of disease detection and severity analysis

Fig. 9 shows the overall flow chart of the proposed method.

5. Experimental results and analysis

The experiments are performed using a computer with NVIDIA GTX1080 GPU, AMD 2600 CPU, and 16G memory. The current study adopts the method of transfer learning to solve the small dataset problem of diseased leaf detection experiments. A pre-trained model of ResNet-101 based on the ImageNet dataset is used to initialize the networks [37, 38]. Table 6 shows the training parameters of CNN.

5.1. Diseased leaf detection results

Fig. 10 shows the detection results of different methods for diseased leaves, where panel (a) depicts the detection results using HOG

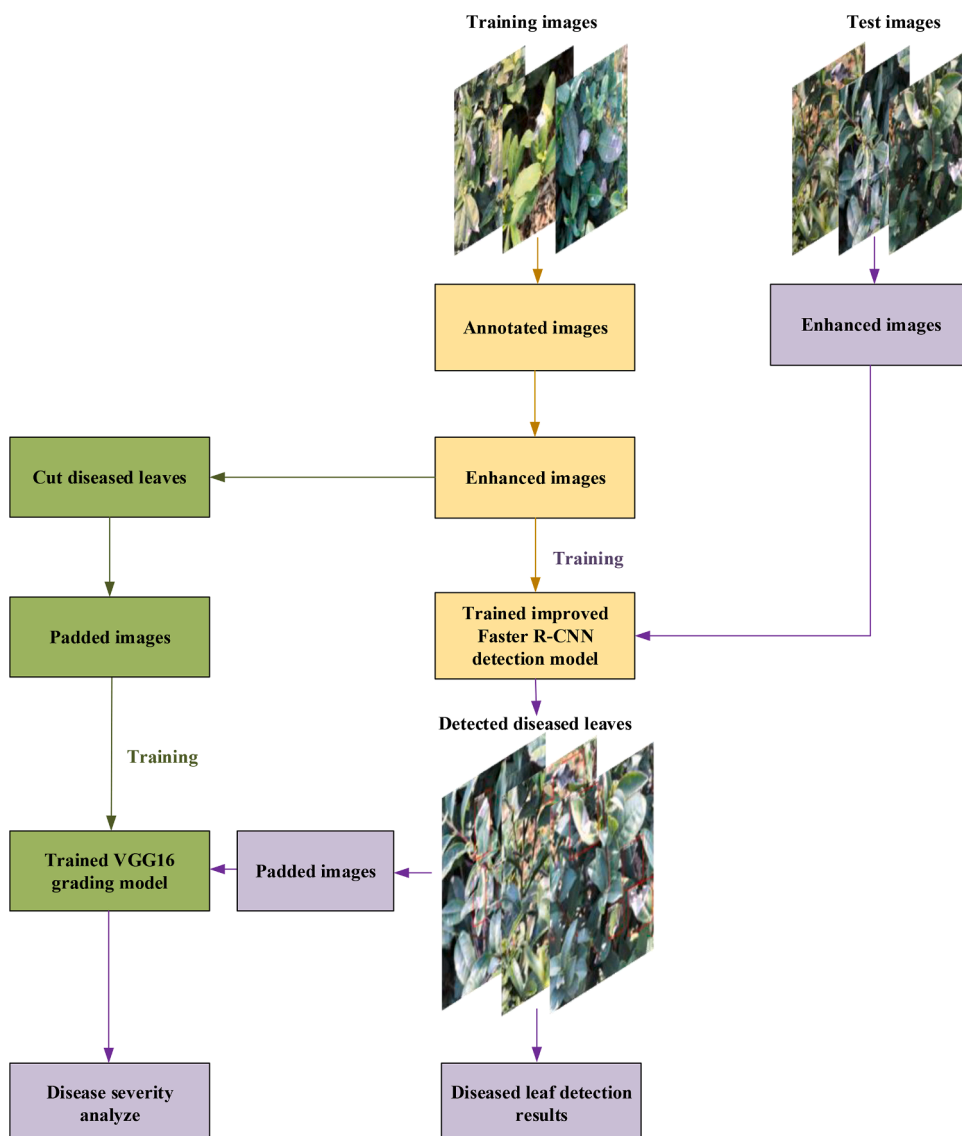


Fig. 9. Overall flow chart of the proposed method.

Table 6
Training parameters of CNN.

Model	Batch size	Iteration	Learning rate	Optimizer	Momentum
YOLO V3	64	10,000	0.001	Momentum	0.9
Faster R-CNN	1	40,000	0.001	Momentum	0.9
Faster R-CNN + FPN	1	40,000	0.001	Momentum	0.9
NAS-FPN-ResNet101	1	40,000	0.001	Momentum	0.9
VGG16	16	20,000	0.0001	SGD	–
Proposed detection method	1	40,000	0.001	Momentum	0.9

feature and SVM classifier, panel (b) shows the detection results of the original Faster R-CNN, and panel (c) illustrates the detection results of the proposed method. The red boxes present the detection results of different methods, and the yellow ones show the diseased leaves with missed detection of TLB. In Fig. 10, the HOG feature and SVM classifier have poor detection effects on the leaves with TLB captured in natural scenes. This method has a low detection accuracy and a high misdetection rate. In addition, the original Faster R-CNN method has limited detection ability for multiscale leaves and leaf occlusion. Hence, the proposed detection method can detect most of the diseased leaves.

The AP value is used to evaluate the performance of different detection methods to quantitatively evaluate the effectiveness of the proposed method in TLB leaf detection [39, 40]. AP is defined as follows:

$$AP = \int_0^1 P(r)dr, \quad (4)$$

where $P(r)$ is the P–R curve, that is, the curve plotted by different recall values and their corresponding precision value. Both values are defined as follows:

$$Precision(P) = \frac{TP}{TP + FP}, \quad (5)$$

$$Recall(R) = \frac{TP}{TP + FN}, \quad (6)$$

where TP, FP, and FN are the number of true positive, false positive, and false negative of the detection results, respectively.

Table 7 shows the Recall, Precision and AP values of detection results using SVM, YOLO V3, NAS-FPN, a series of Faster R-CNN, and the proposed methods.

Table 7 shows that the Recall, Precision and AP values of detection results using the NAS-FPN, the Faster R-CNN series, and the proposed method are higher than those of YOLO V3 and SVM methods. Increasing the convolution to 5×5 or 7×7 improves the effect of extracting the candidate regions compared with the original RPN structure that uses the 3×3 convolution. The Recall, Precision and AP values of detection results using the proposed method is the best, and its Recall, Precision and AP value reaches 93.92%, 95.74% and 91.22%.

Considering that the images were captured in natural scenes and the number of training samples is insufficient, CNN does not deal with light variation and shadow well. Hence, the current study used the Retinex algorithm to enhance the images and reduce the influence of light variation and shadow. Table 7 shows that using Retinex to enhance the image can improve the AP values of detection results.

5.2. Disease severity grading results

Values of Recall, Precision, and Accuracy are used to evaluate the performance of different grading methods to quantitatively evaluate the effectiveness of the proposed method in TLB severity grading. Accuracy is defined as follows:

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN}. \quad (7)$$

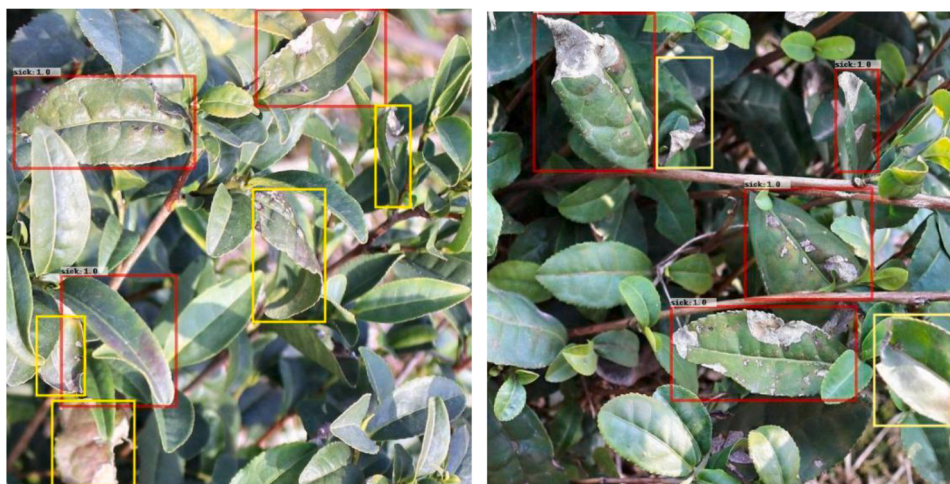
where TN is the number of true negative of the disease severity grading results. Figs. 11–13 show the Recall, Precision, and Accuracy of disease grading of TLB using different methods.

Figs. 11–13 show that the VGG16, the Faster R-CNN, and the proposed method have better disease grading performance than the YOLO V3 and traditional machine learning methods. The disease grading accuracy of the proposed method is more than 9% higher than those of other methods.

Fig. 14 shows two results of disease detection and severity grading using the proposed method. In Fig. 14, the red, green, and blue boxes represent the severe, mild, and missed detection of TLB leaves, respectively. Fig. 15 shows the statistical results of the disease severity of all the test TLB images. Statistics on the number of leaves with different disease severities can help tea farmers analyze the disease severity of tea plants to prevent and control diseases through rational drug use.



(a) Detection results of HOG feature and SVM classifier



(b) Detection results of Faster R-CNN



(c) Detection results of the proposed method

(caption on next page)

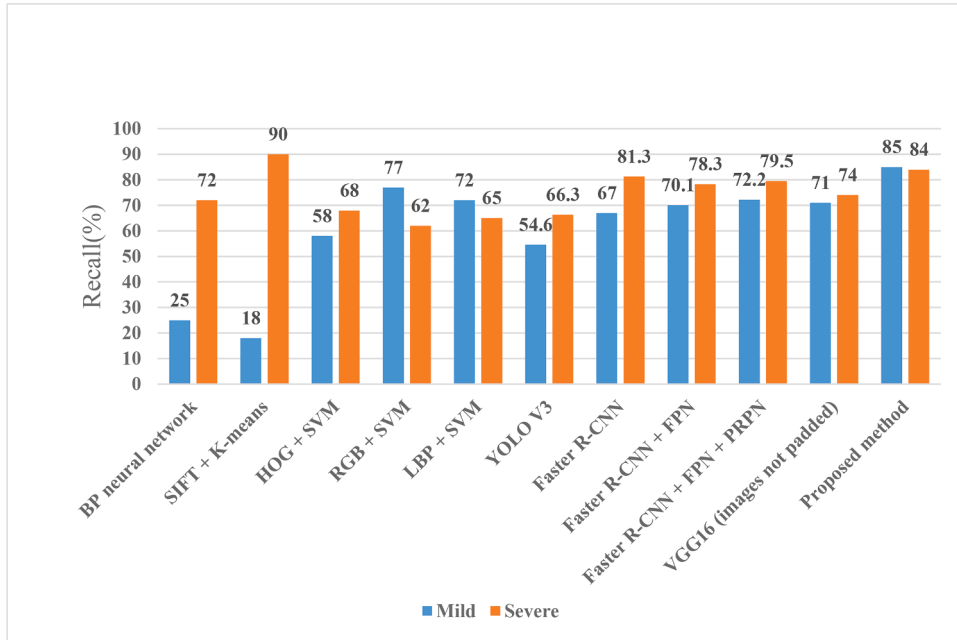
Fig. 10. Detection results of different methods.

- (a) Detection results of HOG feature and SVM classifier;
 (b) Detection results of Faster R-CNN;
 (c) Detection results of the proposed method.

Table 7

Recall, Precision and AP values of detection results using different methods.

Method	Recall (%)	Precision (%)	AP (%)
HOG + SVM[41]	22.05	33.92	11.98
RGB + SVM[42]	23.57	37.13	13.16
LBP + SVM[42]	29.66	40.96	18.63
YOLO V3[43] (images not enhanced)	84.03	87.35	68.21
YOLO V3[43]	85.93	88.98	70.01
Faster R-CNN[32] (images not enhanced)	90.11	92.22	83.72
Faster R-CNN[32]	90.87	92.64	85.04
NAS-FPN-ResNet101[44] (images not enhanced)	90.11	92.59	84.54
NAS-FPN-ResNet101[44]	91.25	92.66	85.52
Faster R-CNN + FPN[33] (images not enhanced)	92.02	93.44	86.97
Faster R-CNN + FPN[33]	92.78	93.85	89.24
Faster R-CNN + FPN (RPN with 5×5 convolution)	93.16	94.59	90.52
Faster R-CNN + FPN (RPN with 7×7 convolution)	92.78	94.21	90.12
Proposed detection method	93.92	95.74	91.22

**Fig. 11..** Recall of disease grading of TLB using different methods.

6. Discussion

The images in the experiments are all captured in natural scenes. The backgrounds of the images are complex, which include dry grass, branches, soil, and so on. TLB spots in the images are similar in color and texture to the background. In addition, the images have uneven lighting and shadows. Traditional machine learning algorithms cannot extract the essential features of diseased leaves, and the detection performance is very limited due to the limitations of manual feature extraction.

Deep learning methods can automatically extract the features of objects. Compared with traditional machine learning methods, deep learning methods have better effects for object detection. NAS-FPN is one of the deep learning networks in the field of object detection, which has achieved remarkable performance on the COCO dataset [44]. NAS-FPN is suitable for a large dataset with hundreds of thousands of images, such as COCO. For a dataset with hundreds of images in the current study, NAS-FPN cannot be fully learned.

The advantage of YOLO V3 lies in its fast model training speed, but its detection performance for small targets is lower than that of

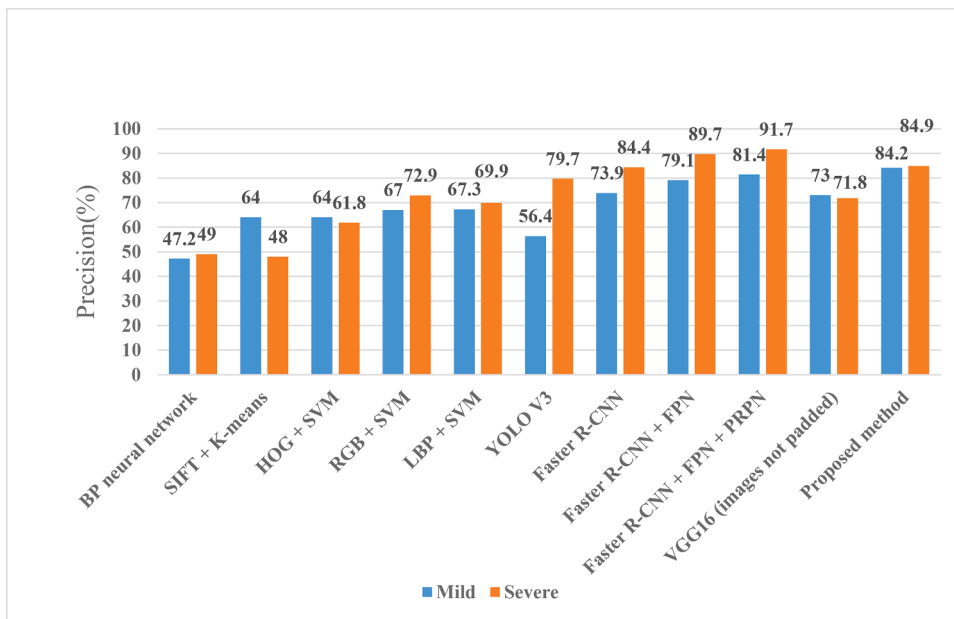


Fig. 12.. Precision of disease grading of TLB using different methods.

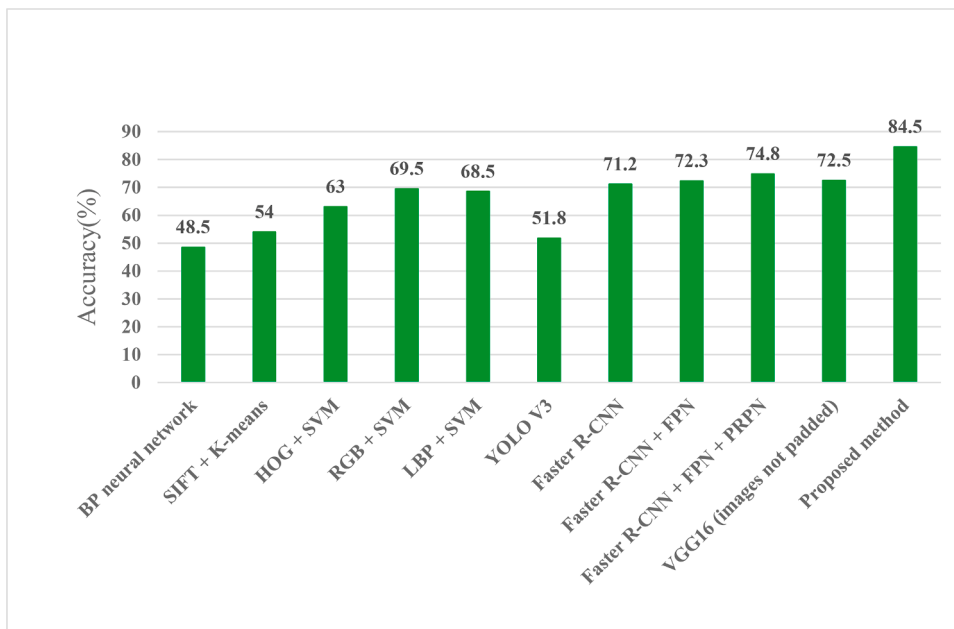


Fig. 13.. Accuracy of disease grading of TLB using different methods.

Faster R-CNN deep learning networks [43]. In particular, with the different severity of TLB, the color, shape, and scale of the diseased tea leaves will change. The performance of YOLO V3 is not as good as the proposed method.

In the current study, the detection of diseased leaves is based on the improved Faster R-CNN. Training Faster R-CNN model takes several hours, but the test time only needs a few seconds. Table 8 shows the training time and test frames per second (FPS) of different deep learning models for diseased leaf detection in this study. The PRPN in the proposed method uses 3×3 and 5×5 convolutions. The number of parameters of 3×3 and 5×5 convolutions is less than that of 7×7 . In Table 8, the training time of the proposed method is less than that of Faster R-CNN + FPN (RPN with 7×7 conv). Compared with other methods, the proposed method can detect diseased leaves more accurately and can provide better technical support for subsequent disease severity estimation and automatic spraying.

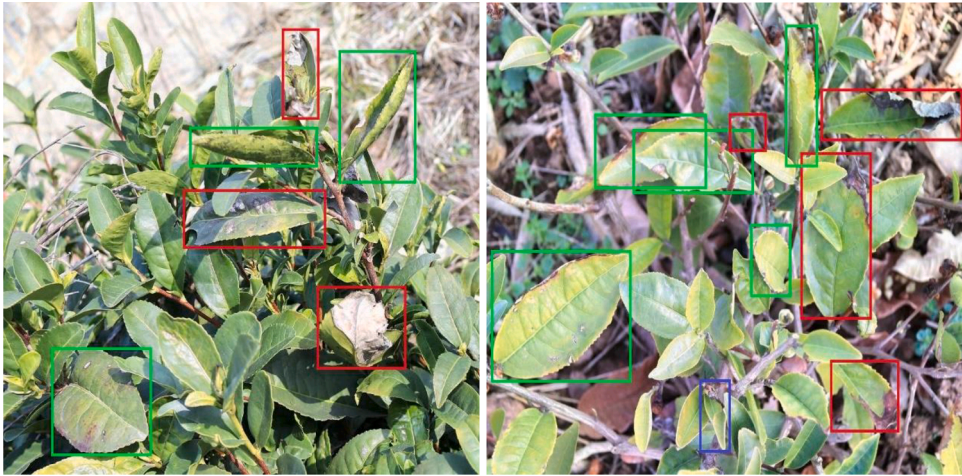


Fig. 14.. Two results of disease detection and severity grading using the proposed method.

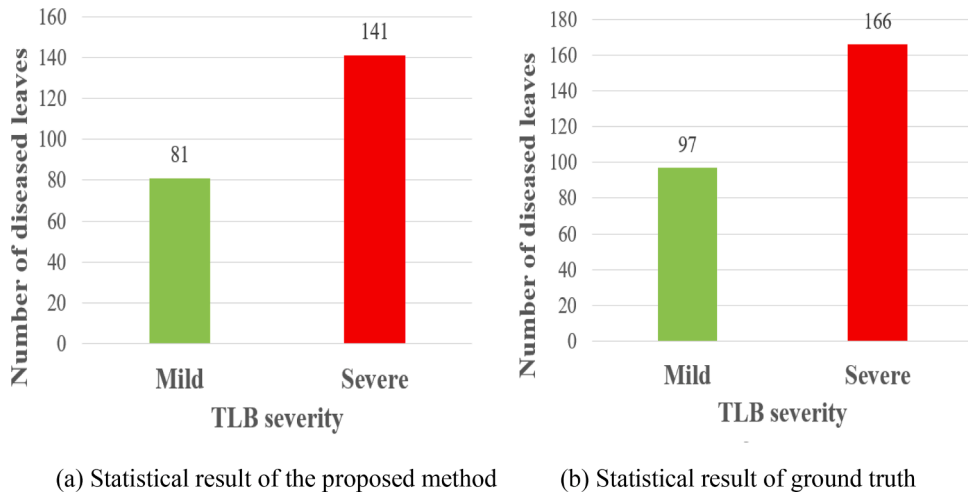


Fig. 15.. Statistical results of disease severity of all the test TLB images.
(a) Statistical result of the proposed method; (b) Statistical result of ground truth.

Table 8

Training time and test FPS of different models for diseased leaf detection.

Model	Training time (minutes)	Test FPS
YOLO V3	183	4.5
Faster R-CNN	270	3.8
NAS-FPN	512	1.6
Faster R-CNN + FPN	293	3.8
Faster R-CNN + FPN (RPN with 5×5 conv)	328	3.7
Faster R-CNN + FPN (RPN with 7×7 conv)	398	3.6
Proposed detection method	340	3.7

Considering that a few diseased leaves are between mild and severe, distinguishing them is difficult, which brings a certain challenge to the disease severity grading model. Therefore, the accuracy of the proposed method in the current study cannot reach 90%. Object detection methods, such as Faster R-CNN and YOLO V3, can complete detection and classification tasks. However, the imbalance in the number of images with different severity of TLB reduces the performance of these object detection methods. The proposed method reduces the impact of the imbalance in the number of images and improves the accuracy of severity grading through image augmentation and separate detection and classification tasks.

The proposed method can simultaneously complete the detection of diseased leaves and the estimation of disease severity. This

method can also perform the detection of diseased leaves or the estimation of disease severity separately according to the needs of tea farmers or spraying robots.

7. Conclusion and future work

TLB is a common tea disease. The current study proposes a deep learning-based method to improve the detection performance and analysis of TLB. The Retinex algorithm is used to enhance the images and reduce the influence of light variation. The Faster R-CNN model is improved on the basis of the characteristics of TLB. The proposed PRPN structure combined with the FPN immensely improves the effect of multiscale detection. The K-means clustering algorithm is used to calculate and modify the anchor ratio for making the detection box consistent with the shape characteristics of leaves infected with TLB. The VGG16 deep learning model is used to accurately classify the severity of TLB, which is beneficial to rational drug use. The experimental results show that the proposed detection and severity analysis model of TLB has remarkable detection and classification performance.

The current study aims to detect and grade the severity of TLB in the natural background. However, TLB is only one of the most common diseases of tea, and other diseases are harming the tea garden, causing great losses to the tea farmers. Hence, these diseases need to be detected, identified, and estimated the severity. After that, further work will continue in this direction.

Author statement

Manuscript title: Detection and severity analysis of tea leaf blight based on deep learning

Gensheng Hu has made significant contributions to data collection and critically revised the organization, structure, and knowledge content of the work.

Huaiyu Wang has drafted the work and made substantial contributions to the conception and design of the work.

Yan Zhang has contributed to data acquisition, analysis and given guidance on the work.

Mingzhu Wan has made an important contribution to the grammar, wording and revision of the manuscript.

All authors read and contributed to the manuscript.

All authors agree to be accountable for the aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

Declaration of Competing Interest

The authors declare that they have no conflicts of interest.

Acknowledgments

The authors thank the Major Natural Science Research Projects in Colleges and Universities of Anhui Province under Grant KJ2020ZD03 and the Open Research Fund of National Engineering Research Center for Agro-Ecological Big Data Analysis & Application of Anhui University under Grant (AE201902) for their support.

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